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ORIGINAL PAPER

An ultrasensitive electrochemical sensor based on cotton carbon fiber composites for the determination of superoxide anion release from cells

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Abstract
A sensor is described for determination of superoxide anion ($O_2^{\cdot -}$). The electrode consists of nitrogen-doped cotton carbon fiber (NCFs) modified with silver nanoparticles (AgNPs) which have excellent catalytic capability. The resulting sensor, best operated at working potentials around −0.5 V (vs. SCE), can detect $O_2^{\cdot -}$ over an extraordinarily wide range that covers 10 orders of magnitude, and the detection limit is 2.32 ± 0.07 fM. The electrode enables the release of $O_2^{\cdot -}$ from living cells under normal or under oxidative stress conditions to be determined. The ability to scavenge the superoxide anions of antioxidants was also investigated. In the authors' perception, the method represents a viable tool for studying diseases related to oxidative stress.

Keywords Cotton carbon fiber · Silver nanoparticles · Superoxide anion · Electrochemical sensor · Antioxidant · Oxidative stress · Cancer cell

Introduction

Oxidative stress is an important carcinogenic factor because it produces of high levels of reactive oxygen species (ROS) that overwhelm cellular repairing systems [1]. Those high levels of ROS species cause damage to DNA and proteins, and promote genomic instability. Therefore, ROS species are becoming increasingly important as risk markers [2, 3]. It is also a challenge to develop effect analytical methods for monitoring the production of ROS under physiological conditions [4]. As one of the most active ROS, the superoxide anion ($O_2^{\bullet -}$) involves in many physiological and pathological processes [5]. Over the last decades, a variety of techniques have been developed for reliable detection of $O_2^{\bullet -}$ including electron spin resonance trapping, fluorescence, spectrophotometry, chromatograph, and chemiluminescence [6, 7]. In comparison to indirect measurements, electrochemical method is the most promising method for $O_2^{\bullet -}$ detection due to its advantages of simplicity, low instrument cost, and feasibility for in-situ monitoring [8]. Presently, it is reported that some nanomaterials (such as Co_3O_4, $Co_3(PO_4)_2$, ZnO, MoS_2, $Mn_3(PO_4)_2$, AuNPs, carbon nanotubes, and graphene) are used to the design of novel interfaces because of their outstanding superoxide dismutase activity and high conductivity [9, 10]. However, they are usually synthesized by chemical method, which is generally complex, time-consuming and environment unfriendly process.

In recent years, conforming to green economy and sustainable development of society, biomass-derived carbon fiber (CFs) has received considerable attention. Compared to inorganic and synthetic organic products, natural products are abundant, renewable, biodegradable, low in cost, and suitable for large-scale production and practical applications [11]. Notably, biomass-derived carbon fiber has its advantages in the development of electrochemical sensors, such as good adsorption performance, large specific surface area, and no agglomeration [12, 13]. However, CFs usually needs special activation and modification to achieve high performance in sensors. So far, many efforts have been made to enhance the performance through modifying the surface property. One efficient way is chemical activation by KOH, $ZnCl_2$, and K_2CO_3 [14–16]. Another is doping hetero atoms or introducing surface functional groups. Among them, nitrogen doping

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has been widely studied to modify the surface property of CFs [17, 18].

In this article, we report the successful preparation of nitrogen-doped cotton carbon fiber (NCF) through carbon thermal reduction method. Then, the Ag nanoparticles (NPs) were loaded on NCF by electrodepositing. To the best of our knowledge, little attention has been paid to the application of cotton carbon fiber on electrochemical sensors (superoxide anion sensors). The developed AgNP/NCF/GCE exhibited outstanding performance towards $O_2^{\bullet-}$ with wide linear range and super low detection limit. More importantly, the constructed electrode was further employed for in situ monitoring of $O_2^{\bullet-}$ released from Glioma cells (U87) in different conditions. The details of the strategy are displayed in Scheme 1.

Material and methods

Apparatus

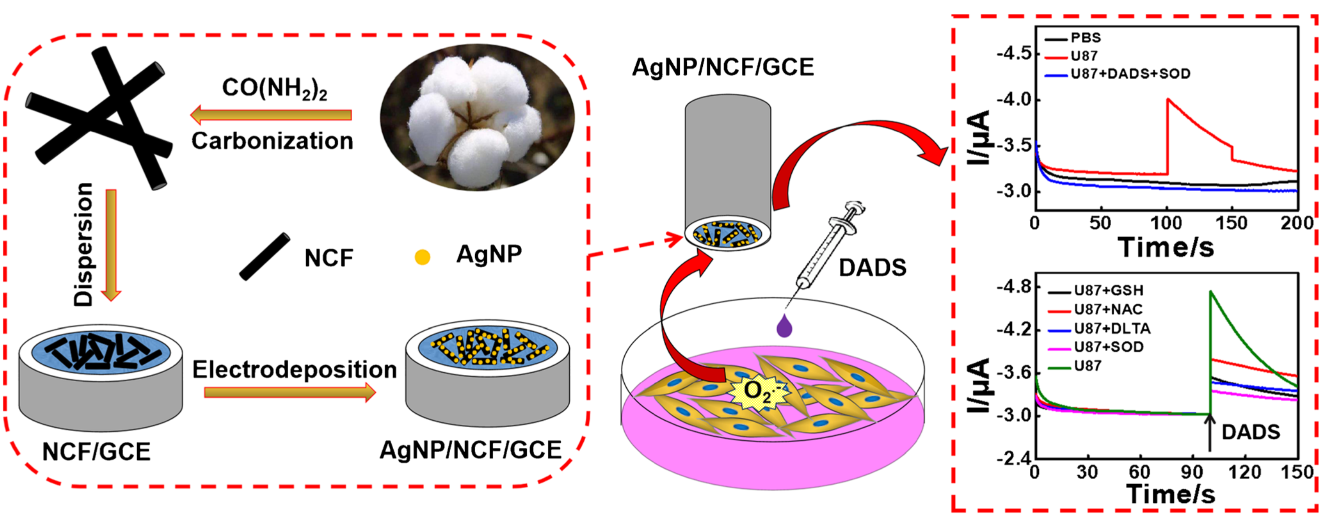
A scanning electron microscope (SEM, Zeiss, Oberkochen, Germany) was utilized to characterize the surface morphology of AgNP/NCF composite. The transmission electron microscopy (TEM) images were obtained from JEM-3010 transmission electron microscope (JEOL Co., Ltd., Japan), and Raman spectra were gotten by an InVia Reflex Raman microscope (excitation-beam wavelength = 632.8 nm). Ultraviolet-visible (UV-vis) spectrometer in the range of 220–400 nm was used for UV-vis absorption studies. All electrochemical experiments in this work were carried out at room temperature on a CHI660D workstation (Austin, TX, USA), which equipped with a conventional three-electrode system consisting of a working electrode, a counter electrode, and a reference electrode. Here, the working electrode is a bare glassy carbon electrode or a modified electrode. The auxiliary electrode is a platinum electrode and a saturated calomel electrode (SCE) as the reference electrode.

Reagents

The degreasing cotton was obtained from Henan Piaoan Group Co., Ltd. (https://www.piaoan.com). Urea was provided by Sinopharm Chemical Reagent Co., Ltd. (https://www.sinoreagent.com); KO_2, diallyl disulfide (DADS) and N-acetyl-L-Cysteine (NAC) were purchased from Aladdin reagent(https://www.aladdin-e.com). DL-Thioctic acid (DLTA), reduced glutathione (GSH) and superoxide dismutase (SOD) were provided by Shanghai yuanye Bio-Technology Co., Ltd. (https://www.shyuanye.com). DMSO, 18-crown-6, and 4 Å molecular sieves were purchased from Beijing Chemical Works (https://www.beijingchemworks.com). $AgNO_3$ and KNO_3 came from Tianjin Guangfu Fine Chemical Research Institute (https://www.guangfu-chem.com). Phosphate buffer solution (PBS, pH 7.0) was prepared by mixing suitable amounts of 0.2 M NaH_2PO_4, Na_2HPO_4. A physiological PBS solution was prepared for washing U87 cells mainly using KH_2PO_4 (1.76 mM), Na_2HPO_4 (10.14 mM), NaCl (136.75 mM) and KCl (2.28 mM). All other chemicals were of analytical grade and the solutions used were all prepared by double distilled water.

Preparation of nitrogen-doped cotton carbon fiber (NCFs) composites

NCFs were synthesized by a two-step process. In the first step, 1.0 g of cotton was torn into pieces and soaked in 50 mL ultrapure water containing 6.0 g of urea. Then the cotton was sonicated for approximately 3 h to allow the cotton to disperse evenly in the urea solution. Finally, the cotton was dried in a vacuum oven at 80 °C for 24 h. In the second step,



Scheme 1 Schematic illustration of the procedure for the preparation of the NCF-based electrochemical sensor and application for detection of superoxide anion released from cells

the vacuum-dried cotton was directly carbonized at 800 °C with a heating rate of 2 °C min^{-1} for 2 h under a nitrogen atmosphere, and formed nitrogen-doped cotton carbon fiber (NCFs). (The sufficiently detailed preparation of nitrogen-doped cotton carbon fiber (NCF) composites is shown in supporting information).

Preparation of the $O_2^{\bullet-}$ sensor

A bare glassy carbon electrode (GCE) was first polished with 0.3 and 0.05 μm alumina slurry to a mirror-like, respectively. Next NCF aqueous solution (7.0 μL) was dispensed onto the surface of GCE and dried to obtain NCF/GCE. Then, the modified electrode was used as a working electrode for electrodepositing silver, which expressed as AgNP/NCF/GCE electrode. The electrodeposition conditions of silver nanoparticles are as follows: time of AgNP electrodeposition (240 s); potential of AgNP electrodeposition (0.0 V) and $AgNO_3$ concentration (1.0 mM).

Synthesis of superoxide anion solutions

The chemical generation of $O_2^{\bullet-}$ was performed according to the ref. [19]. The dimethyl sulfoxide (DMSO) solution was dehydrated first with 4 Å molecular sieves. Then KO_2 and 18-crown-6 were dissolved above DMSO solution. And next the solution was sonicated for 10 min to obtain the superoxide anion aqueous solution. Finally, based on the molar absorptivity of $O_2^{\bullet-}$ in DMSO (2006 M^{-1} cm^{-1} at 271 nm), we estimated the concentration of $O_2^{\bullet-}$ by UV-Vis spectrometer.

Culture of glioma cells

Glioma cells (U87) came from Chinese Academy of Medical Sciences (Institute of Hematology). They were cultured in DMEM supplemented containing 10% fetal bovine serum (FBS), penicillin (100 U mL^{-1}), and streptomycin (100 U mL^{-1}) in a humidified atmosphere containing CO_2 (5%) at 37 °C. Penicillin and streptomycin were obtained from Sigma (USA).

Electrochemical detection of superoxide anion released from cells

Glioma cells (U87) were seeded in a 6-well plate (1.5×10^5 cells per well) for 24 h and used for electrochemical experiments. Prior to each electrochemical experiment, the cell nutrient solution was removed by pipetting and then washed twice with the pretreated physiological PBS solution (pH 7.4). Finally, 3.0 mL of phosphate buffer was added to each well for experimental measurements. The current response of $O_2^{\bullet-}$ released from U87 cells was detected using AgNP/NCF/GCE as working electrode. When the response of the superoxide anion reached a stable level, diallyl disulfide (DADS) was injected as a stimulant into the cell suspension to induce the U87 cells to produce $O_2^{\bullet-}$. In addition, DADS was added to the phosphate buffer in the absence of U87 cells and the measured current response was used for comparison with previous experiments.

Results and discussion

Morphological characterization of AgNP/NCF/GCE

Scanning electron microscopy (SEM), transmission electron microscopy (TEM) and Raman spectra characterization are conducted for studying the unique morphology of the as-prepared AgNP/NCF and comparison samples (NCF). As shown in Fig. 1a, the surface of nitrogen-doped cotton carbon fiber is slightly smooth, and the longitudinal distribution of the surface has obviously trench, and the diameter of cotton carbon fiber is about 5~8 μm. The energy dispersive X-ray spectroscopy (EDX) of NCF/GCE can be seen in Fig. S1. The strong peaks for C, N and O demonstrate that NCF composites have been synthesized. Furthermore, when AgNP electrodeposited on NCF composites (Fig. 1b), one can see that high density AgNP are disorderly loaded on the surface of NCF. Fig. 1c and d are the TEM images, where a large number of small-sized AgNP are uniformly distributed on NCF with average size of 15~20 nm. The AgNP on the surface of NCF is further confirmed by high-resolution transmission electron microscopy (HRTEM) (Fig. 1e). The measured fringe lattice of AgNP corresponding to the Ag (111) plane is found to be 0.236 nm, and the particle size is in agree with the Fig. 1c and d. The Raman spectrum of the CFs and NCF can be seen from Fig. 1f. The broad peaks located at about 1354 cm^{-1} and 1595 cm^{-1} are assigned to the D band (disordered carbon) and G band (graphitic carbon) of carbon materials, respectively. The intensity ratio of the two peaks (I_D/I_G) is used to evaluate the disorder degree of carbon materials. Compared with CFs ($I_D/I_G = 0.95$), NCFs have a higher I_D/I_G intensity ratio ($I_D/I_G = 1.11$), confirming more defects on the nitrogen doped cotton carbon fiber.

Additionally, electrochemical characterization of AgNP/NCF/GCE is performed as shown in Fig. S2 and Fig. S3. The experimental results confirmed that NCF composites are fit for effective load silver nanoparticles, and accelerate the electron transfer of $[Fe(CN)_6]^{3-/4-}$.

Electrochemical response of superoxide anion at AgNP/NCF/GCE

Fig. 2 shows the cyclic voltammetry experiment of different electrodes in the absence and in the presence of 1.0 mM $O_2^{\bullet-}$ in N_2-saturated 0.2 M PBS solution (*pH* 7.0) at a scan rate of 100 mV s^{-1}. Obviously, the well-defined characteristic peaks

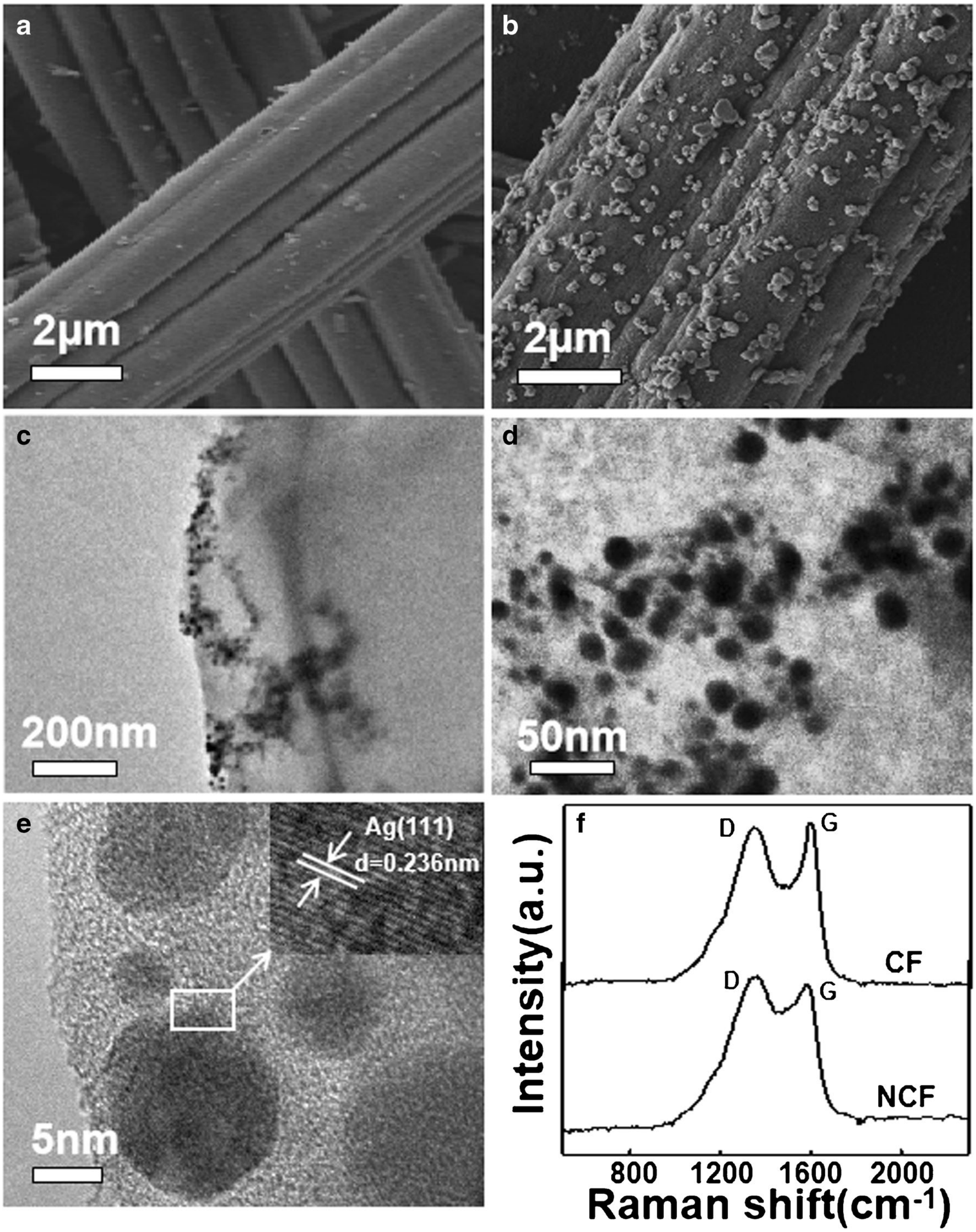


Fig. 1 SEM images of various modified GCE substrates: **a** NCF/GCE, **b** AgNP/NCF/GCE; TEM images of AgNP/NCF/GCE **c** and **d**; **e** HRTEM image of AgNP/NCF/GCE; **f** Raman spectra of CFs and NCF

of $O_2^{\bullet-}$ reduction are observed in each figure, and the responses in curve (b) increased greatly than in curve (a), showing that superoxide anion ($O_2^{\bullet-}$) had good electrochemical responses in the four electrodes. From curve (b) of Fig. 2a and b, small and broad $O_2^{\bullet-}$ reduction peaks appeared on the GCE and NCF/GCE. However, the greater peak currents appeared at the AgNP/GCE and AgNP/NCF/GCE, suggesting that silver nanoparticles have excellent catalytic activity for reduction of $O_2^{\bullet-}$. Comparatively, the response in Fig. 2d is the biggest among the four CV pictures, which is approximately 3.34 times higher than the bare GCE. The excellent current is ascribed to the synergistic amplification effect of the two kinds of AgNP and NCF. Moreover, compared to GCE, NCF/GCE and AgNP/GCE, the peak potential of $O_2^{\bullet-}$

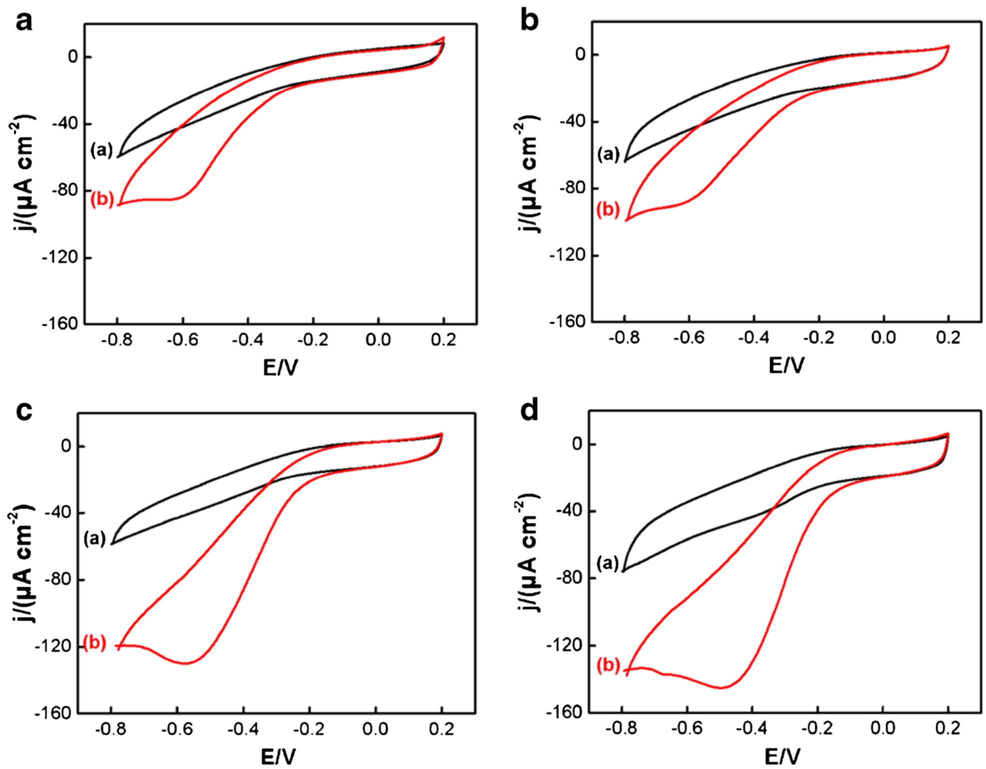


Fig. 2 CVs of different electrodes in the absence (**a**) and the presence of 1.0 mM $O_2^{•-}$ (**b**) in N_2-saturated 0.2 M PBS (pH 7.0): **a** bare GCE, **b** NCF/GCE, **c** AgNP/GCE and **d** AgNP/NCF/GCE, at 100 mV/s

reduction on the AgNP/NCF/GCE is shifted to more positive, and the result further illustrates that AgNP/NCF/GCE shows excellent catalytic ability toward $O_2^{•-}$ [20].

Moreover, we calculated the average electroactive surface areas of bare GCE and NCF/GCE shown in Fig. S4. The area of NCF/GCE (0.1158 cm^2) is 1.45 times larger than bare GCE (0.0801 cm^2), further suggesting that NCF composites have larger effective area to load AgNP greatly.

Optimization of experimental variables

We optimized the experimental variables to get the best performance of the sensor. The electrochemical experimental results are shown in Fig. S5. One can see that the as-prepared sensor had the best responses for $O_2^{•-}$ under the following conditions: mass ratio of Urea and Cotton (5:1); NCF volume (7.0 μL); NCF concentration (0.5 mg mL^{-1}); time of AgNP electrodeposition (240 s); potential of AgNP electrodeposition (0.0 V) and $AgNO_3$ concentration (1.0 mM).

Calibration plots of superoxide anion on the AgNP/NCF/GCE

Under above optimal conditions, linear sweep voltammetry (LSV) is used for measurement of $O_2^{•-}$ concentration quantitatively. Fig. 3a and c are the typical LSV curves for $O_2^{•-}$ responses on the AgNP/NCF/GCE. One can see that the current responses increased along with the concentrations of $O_2^{•-}$ in the ranges from 6.96×10^{-15} to 7.22×10^{-5} M shown in Fig. S6. Fig. 3b and d are the corresponding calibration plots, and their linear regression equations are Ip(μA) = 3.9568 g lg[$O_2^{•-}$] (μM) + 49.132 ($R_2 = 0.9943$) ($6.96 \times 10^{-15} \sim 7.37 \times 10^{-8}$ M), and Ip(μA) = 0.2649 [$O_2^{•-}$] (μM) + 48.368 ($R_2 = 0.9932$) ($3.57 \times 10^{-7} \sim 7.22 \times 10^{-5}$ M), respectively. The limit of detection (*LOD*) is 2.32×10^{-15} M at a signal to noise ratio of 3. Such extraordinary performance towards $O_2^{•-}$ is ascribed to the good conductivity of NCF composites and its being used as effective load matrix for the deposition of AgNP. As a control experiment, the LSV responses of the AgNP/GCE towards the $O_2^{•-}$ were shown in Fig. S7, and the linear range is 5 orders of magnitude ($4.99 \times 10^{-10} \sim 1.75 \times 10^{-8}$ M, $5.64 \times 10^{-7} \sim 1.72 \times 10^{-5}$ M) with a limit of detection (*LOD*) of 1.66×10^{-10} M. Thus, NCF also play an indispensable role in the construction of electrochemical sensor in this paper.

Furthermore, compared to the literatures in Table 1, it is clear that the performance of our sensor is superior to other electrochemical sensors reported previously.

Selectivity, stability and reproducibility of the $O_2^{•-}$ sensor

In order to evaluate the selectivity of the constructed electrochemical sensor, various related interfering species in biological system were tested as control experiments including

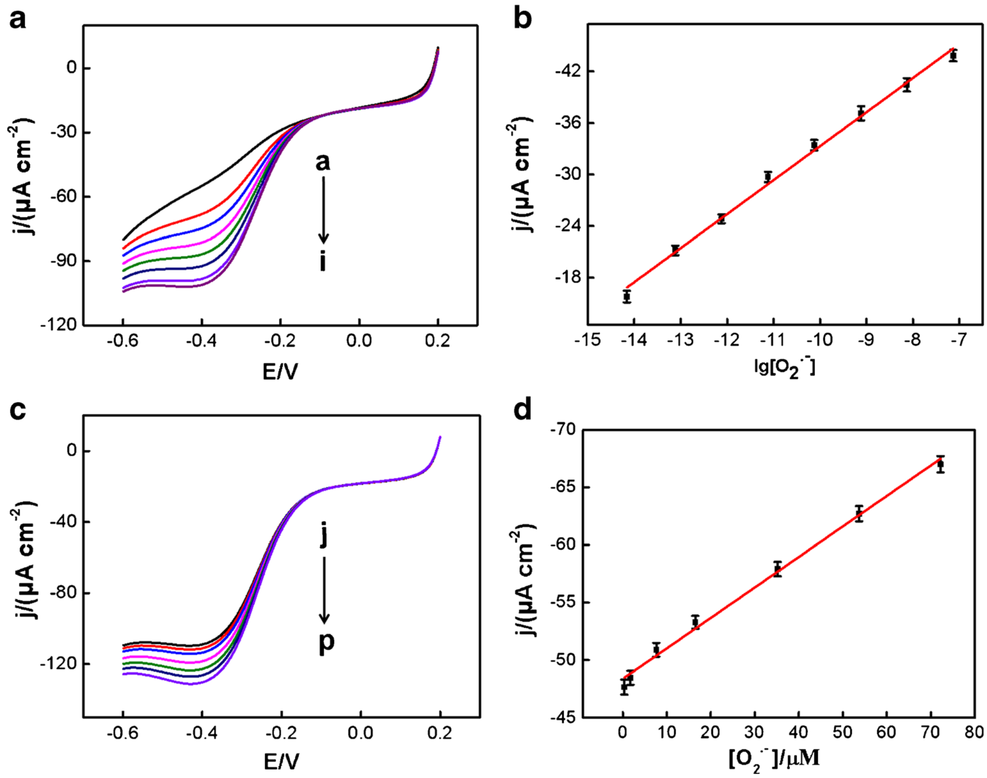


Fig. 3 LSV curves obtained at AgNP/NCF/GCE for the analysis of different concentrations of $O_2^{\bullet-}$ in N_2-saturated 0.2 M PBS (pH 7.0) under continuous stirring. Fig. 3a and c are current responses of $O_2^{\bullet-}$ in the low-concentration (from (a) to (i): 0, 6.96×10^{-15}, 7.60×10^{-14}, 7.62×10^{-13}, 7.57×10^{-12}, 7.52×10^{-11}, 7.66×10^{-10}, 7.42×10^{-9}, 7.37×10^{-8} M) and the high-concentration (from (j) to (p): 3.57×10^{-7}, 1.75×10^{-6}, 7.63×10^{-6}, 1.65×10^{-5}, 3.52×10^{-5}, 5.37×10^{-5}, 7.22×10^{-5} M), respectively; Fig. 3b and d show the corresponding calibration plots of current response between the logarithm of the $O_2^{\bullet-}$ concentration and the concentration of $O_2^{\bullet-}$. Each point is expressed as mean ± standard deviation ($n = 3$), at 100 mV/s

0.5 mM ascorbic acid (AA), uric acid (UA), dopamine (DA), NO_2^-, NO_3^- and $ONOO^-$. Fig. 4 showed the current responses of interfering species on AgNP/NCF/GCE. Compared to the significant current of $O_2^{\bullet-}$, these electrochemical responses of the potentially interfering substances can be negligible, confirming that AgNP/NCF/GCE performed superior selectivity toward $O_2^{\bullet-}$ and had potential capacity for $O_2^{\bullet-}$ detection in complex biological environments.

The reproducibility of five identically fabricated AgNP/NCF/GCE is determined by analyzing its amperometric responses to 1×10^{-5} M $O_2^{\bullet-}$ under the identical conditions. The corresponding performance demonstrates the relative standard deviation (*RSD*) of 3.23% (Fig. S8a). Moreover, we also tested the stability of the modified electrode and the result is obtained from a continuous CV scan for as-constructed AgNP/NCF/GCE in the N_2-saturated 0.2 M PBS (*pH* 7.0). One can see that current still remained 96.6% of the original response in Fig. S8b. These results revealed that the as-prepared AgNP/NCF/GCE exhibited good reproducibility, stability and selectivity for the $O_2^{\bullet-}$ detection.

Investigation the ability to scavenge superoxide anions of antioxidants

As well known, superoxide anions is one of the most vital biomolecules and a potential marker for tumor cells [28]. Thus, it is meaningful that electrochemical testing $O_2^{\bullet-}$ from cancer cells for the real-time with the as-constructed electrode

Table 1 Comparison of various $O_2^{\bullet-}$ sensor

Electrode materials	Detection potential(V)	Linear range (nM)	LOD (nM)	Ref.
FITC@mSiO_2@hmSiO_2@HE		$200\sim2 \times 10^4$	80	[21]
Mn-MPSA-MWCNTs/SPCE	0.7	$0\sim1.817\times 10^3$	127	[22]
SOD/CNT/PEDOT/GCE	−0.3	$2.0\times 10^4\sim3.0\times 10^6$	1000	[23]
SOD/porousPt–Pd/SPCG	−0.1	$1.60\times 10^4\sim1.54\times 10^6$	130	[24]
PDDA/MWCNTs-Pt/GCE	−0.5	$0.70\times 10^3\sim3.00\times 10^6$	100	[25]
$Co_3(PO_4)_2$NRs/GCE	0.7	$5.76\sim5.39\times 10^3$	2.25	[26]
AgNPs/Cys-MWCNTs/GCE	−0.6	$0.07\sim7.41 \times 10^4$	2.33×10^{-2}	[27]
AgNP/NCF/GCE	−0.5	$6.96 \times 10^{-6} \sim 73.7$ $357\sim 7.22 \times 10^4$	2.32×10^{-6}	This work

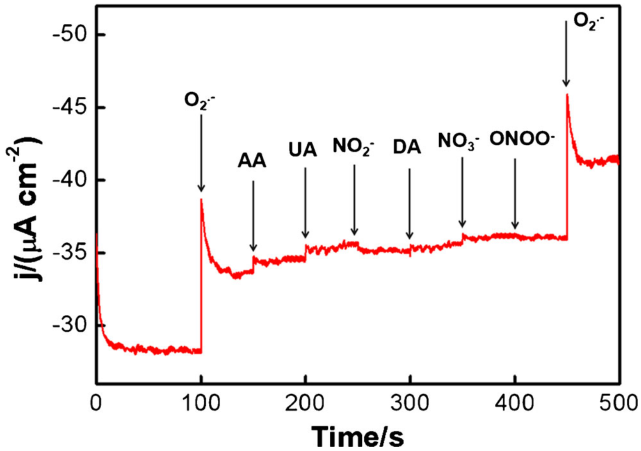


Fig. 4 Amperometric i-t response of AgNP/NCF/GCE upon the addition of various species (0.5 mM each of AA, UA, DA, NO_2^-, NO_3^-, $ONOO^-$) and twice addition of 1×10^{-5} M $O_2^{\bullet-}$ in 0.2 M PBS (pH 7.0) under a stirring condition, at −0.50 V

in this work. Here, Glioma cells (U87) is employed as a kind of model cancer cell, and Diallyl disulfide (DADS) is chosen as a stimulant agent to induce cells producing superoxide anions.

Firstly, we studied the electrochemical responses obtained at the AgNP/NCF/GCE in different conditions. As shown in Fig. 5a, it is clearly that the current of curve (b) will increase with time after 170 s. The result shows that the U87 cells slowly release ROS by themselves, and it can be tested by the sensor described here. As DADS is added the cell solution at about 100 s, the current in curve c increased greatly. To demonstrate whether the increased current signal is caused by the stimulated cells, SOD (300 U/mL) is added to U87 cells at the time of 150 s, which is the most powerful antioxidant against ROS in cell [29]. We found the current decreased immediately because of the decomposition of $O_2^{\bullet-}$ (c). Above experiment results are agree with the other reports.

Secondly, we also studied the influence of DADS dose on the amount of $O_2^{\bullet-}$ producing by U87 cells in Fig. 5b. The current response increased along with the concentrations of DADS (20, 40, 60, 100 μM) and reached a maximum at 60 μM. When DADS were added at 150 s again, the similar

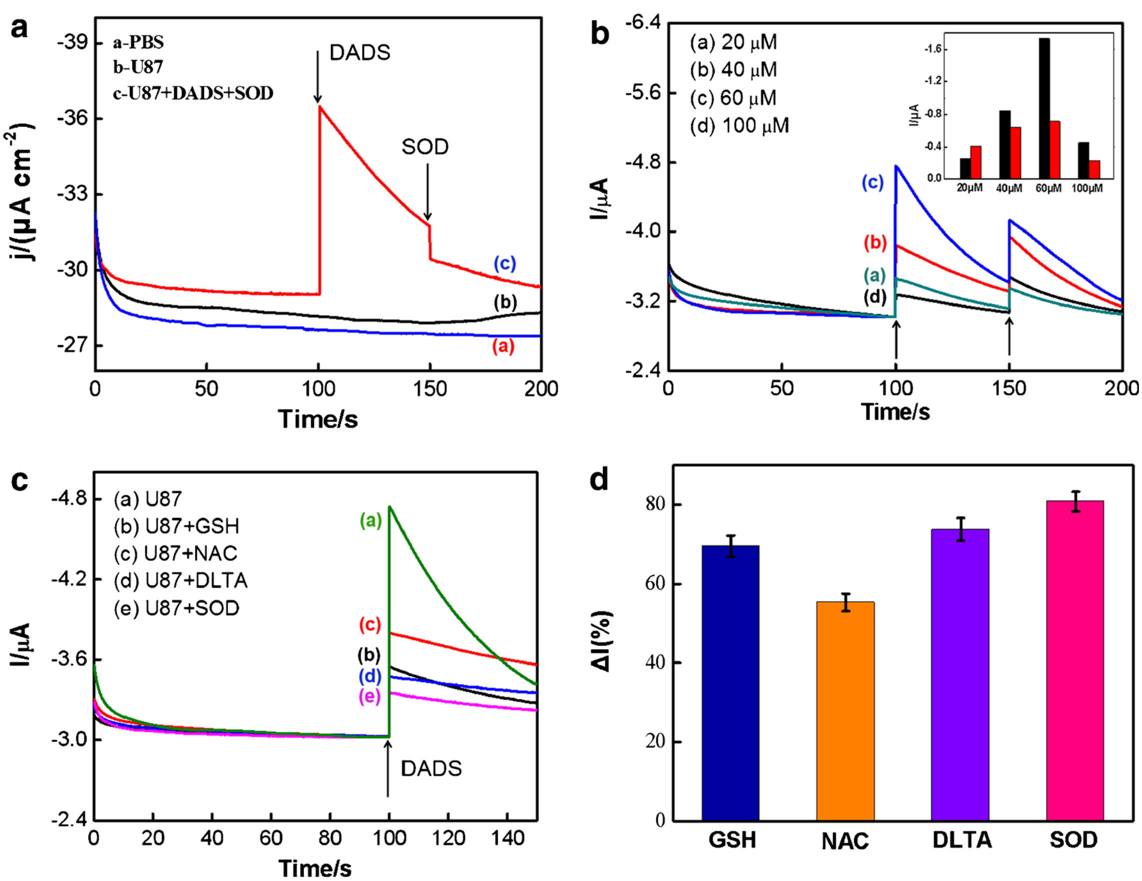


Fig. 5 **a** Time course of the blank PBS (**a**), the U87 cells (**b**), U87 cells upon the addition of 40 μM DADS and 300 U/mL SOD (**c**); **b** Amperometric responses of $O_2^{\bullet-}$ released from U87 cells induced with different concentrations of DADS. Inset is a columnar comparison of corresponding current changes; **c** Current responses of $O_2^{\bullet-}$ released from U87 cells in the absence of antioxidants (**a**) and presence of antioxidants: (**b**) 4 mM reduced glutathione (GSH), (**c**) 4 mM N-acetyl-L-cysteine (NAC), (**d**) 4 mM DL-Thioctic acid (DLTA), (**e**) 300 U/mL SOD, at −0.50 V; **d** The percentage decrease of the current response derived from $O_2^{\bullet-}$ released by the U87 cells in the presence of different antioxidants compared to the absence of antioxidants

responses were also observed. In order to compare the responses obtained from stimulus at 100 and 150 s, a histogram is made as shown in the insert of Fig. 5b. Black column and red column represent current responses at 100 and 150 s, respectively. It's worth noting that the current change is the biggest at 60 μM, and then decreased greatly at 150 s. The possible reason is due to produce excess $O_2^{\bullet-}$ from U87 cells in the first stimulus, which may cause oxidative damage or even kill cells. The electrochemical experimental results will also be proved again by bright-field microscopy images of U87 cells (Fig. S9c). Furthermore, we also evaluated the release amount of $O_2^{\bullet-}$ from the U87 cells. In Fig. 5b, the peak value of current signal (1.734 μA) in curve c indicated the maximum $O_2^{\bullet-}$ efflux, which corresponds to about 9.79×10^{-15} M $O_2^{\bullet-}$ obtained from the linear regression equation. Then, the $O_2^{\bullet-}$ released from each cell is calculated about 1.31×10^{-20} mol·cell^{-1}·s^{-1}, which is smaller than the earlier reports [30]. Here, it should be noted that the electrochemical response obtained at the AgNP/NCF/GCE is the average value of ROS release from living cells in per well (1.5×10^5 cells). It is not the instantaneous value of ROS release from living cells because we cannot control the distance between the electrode and the cells, which is a major limitations of this work.

In order to prevent oxidative stress caused by excess superoxide anion, we further investigated the ability to scavenge superoxide anion of antioxidants. As well known, reduced glutathione (GSH) [31], N-acetyl-L-cysteine (NAC) [32, 33], and DL-Thioctic acid (DLTA) [34] are excellent drug antioxidants. From Fig. 5c, when U87 cells were induced directly with 60 μM DADS, a strong current response is observed in curve (a). In contrast, when above different antioxidants and SOD (300 U/mL) were added to the U87 cells in advance and then 60 μM DADS is added to stimulate the U87 cells, it is clearly that these current responses ((b)-(e)) were significantly lower than that of curve (a). Further data analysis is shown in Fig. 5d, compared with curve a, the current responses of the $O_2^{\bullet-}$ were distinct decreases by 69.6%, 55.3%, 73.8% and 80.8% in curve (b), (c), (d) and (e), respectively, revealing that all these antioxidants have strong ability to scavenge $O_2^{\bullet-}$. Especially, the scavenging capacity of DLTA is basically comparable to the SOD.

As well known, SOD enzyme is the most powerful antioxidant against ROS in cell, and it is indispensable to protect body cells from excessive oxygen radicals and free radicals. However, the levels of SODs will decline with age, whereas free radical formation increases, which may promote aging or cause cell death. Our study finding indicated that in the cell system, the scavenging capacity of some drug antioxidants (GSH, DLTA) is basically comparable to the SOD enzyme, which can counteract the cytotoxicity of ROS in cells via scavenging $O_2^{\bullet-}$. Thus, proper daily drug antioxidants supplementation will protect the immune system, reduce one's chances of diseases significantly and ultimately slow down aging process.

Conclusions

In summary, we synthesized a new type of electrode material from natural products, cotton carbon fiber. The rationally designed AgNP/NCF sensing platform exhibited outstanding electro catalytic activity to the superoxide anion. It offered a linear range of 10 orders of magnitude and a super low detection limit of 2.32 ± 0.07 fM, electrochemical sensitivity was 3.9568 μA·μM^{-1}·cm^{-2}. Due to its outperformed analytical characteristics, the sensor is successfully adapted to detection of $O_2^{\bullet-}$ released from living cells directly. Moreover, the scavenging capacity of 3 drug antioxidants was also studied by electrochemical method. Based on the relationship among ROS, oxidative stress, and related diseases, these results suggested antioxidants can protect the cells against oxidative stress. Our work provides a new strategy to fabricate highly sensitive ROS sensors and to prevention of oxidative stress related diseases.

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Compliance with ethical standards The authors declare that they have no competing interests.

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